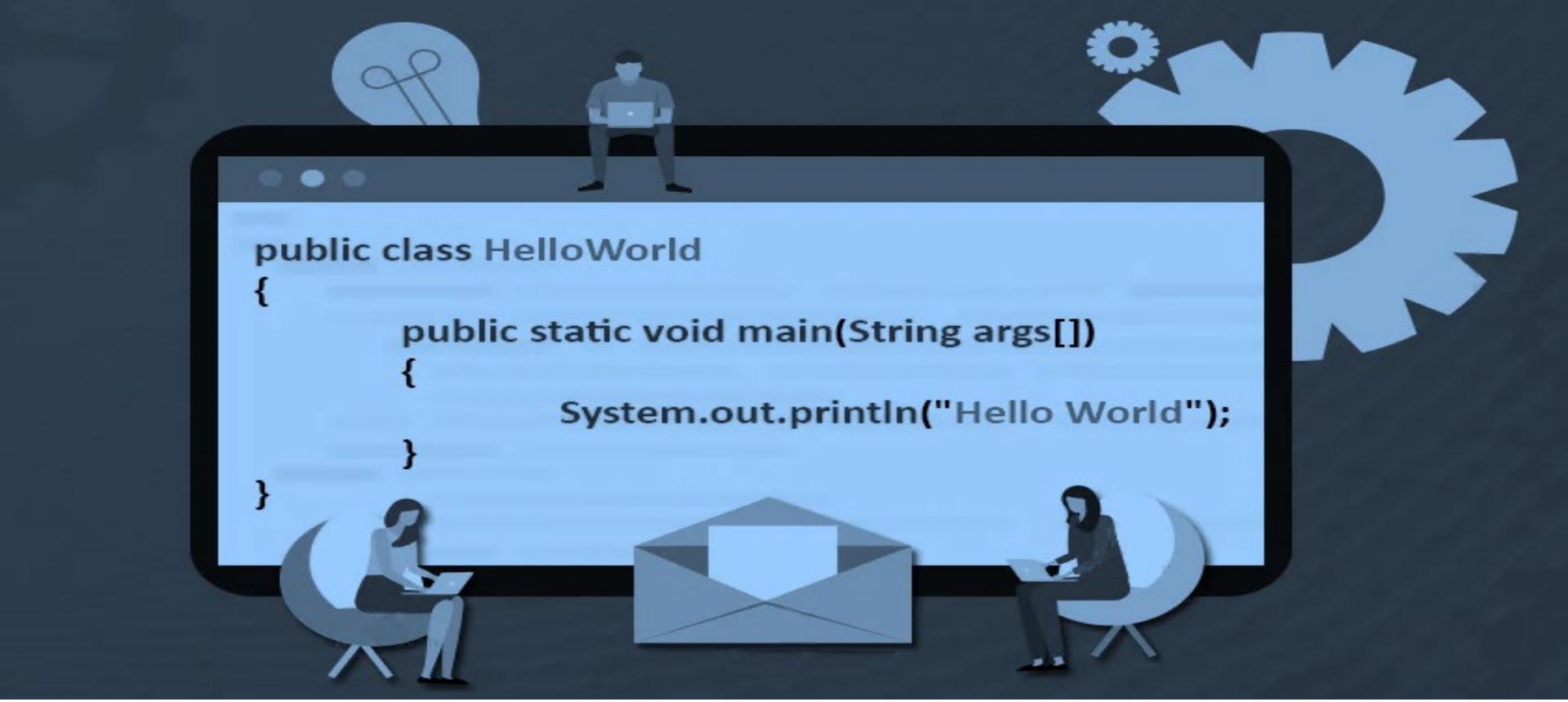

Java Programming

CPT111 – Lecture 1

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CPT111 Java Programming

Lecture 1

Simple Java Program, Data, Variables

Welcome!

- Welcome to Lecture 1 !

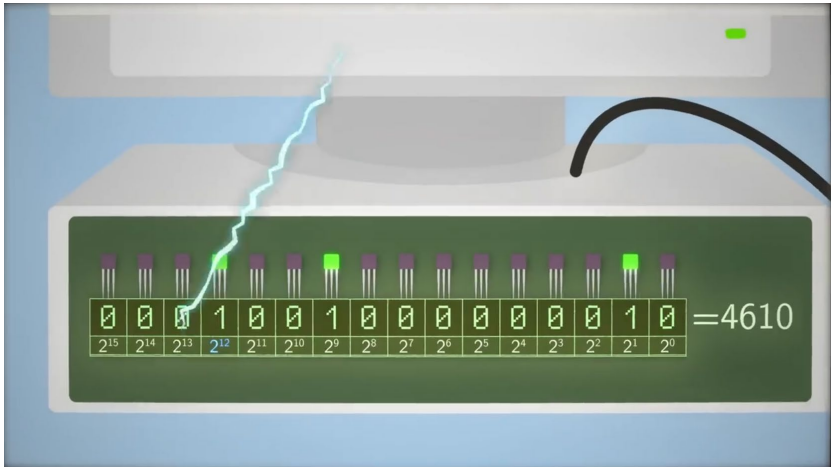
- In this lecture we are going to learn to write and run a simple Java program
  - learn its elements
    - Class
    - Main() method
    - Statements
  - learn about variables
    - Declaration
    - Initialization
    - Assignment

Binary, Bit and Bytes

- To a modern computer, everything is binary
  - 1 bit is 1 binary digit: 0 or 1 ; ON or OFF ; TRUE or FALSE
- Every bit we add doubles the total we can represent
  - 1 bit – 2
  - 2 bits – 4
  - 3 bits – 8
  - 4 bits – 16
  - ...
  - 8 bits – 256 – one Byte
- Mathematically: n bits yields 2^n patterns (2 to the n-th power)
  - https://www.wikihow.com/Convert-from-Decimal-to-Binary

The Universe is Hostile to Computers

- What happened if **one single bit** in a computer got flipped?
- Watch this video from Veritasium:
  - https://box.xjtlu.edu.cn/f/be27c89a55ad43d58c2a/



Bit, Bytes and Storage

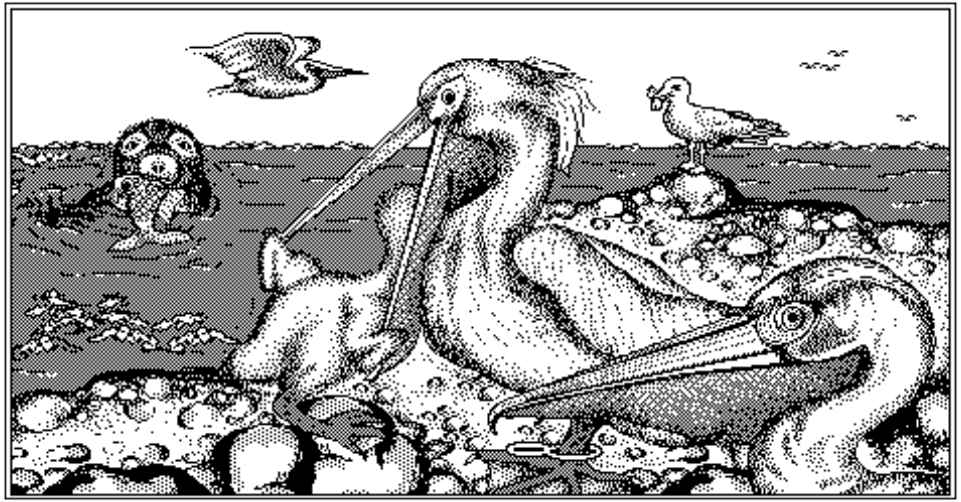
- More bits mean more potential storage
  - Longer instructions, more data
  - Better quality files

- 8 bits = 1 byte ≈ often 1 character, e.g. "a"
- 1024 bytes = 1 kilobyte ≈ several paragraphs
- 1024 kilobytes = 1 Megabyte ≈ about 200 pages of text
- 1024 Megabytes = 1 Gigabyte ≈ 256 MP3 files
- 1024 Gigabytes = 1 Terabyte ≈ 40 large movies

- Next, let's give some examples how things are stored/represented in binary!

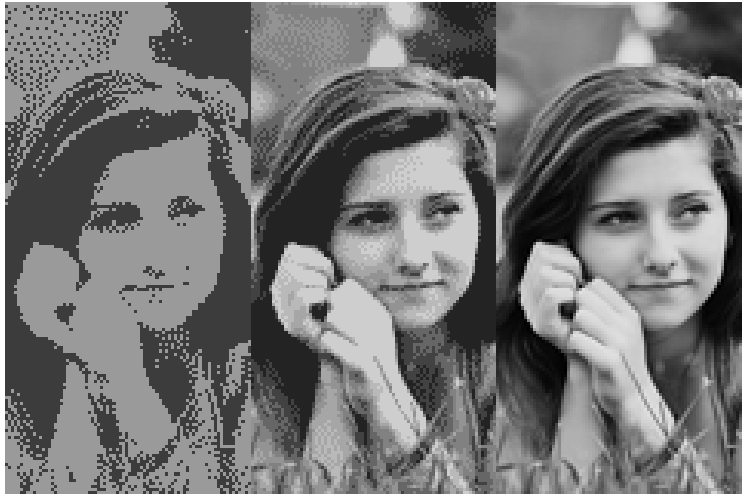
1-Bit Colour

- Black or white
  - with a high enough resolution some pretty good effects could be had by mixing black and white to get grey effects



8-Bit Colour – Grayscale

- 8-bit grayscale gave us decent photos – in "black and white" (which is not!)
  - 256 different colours possible



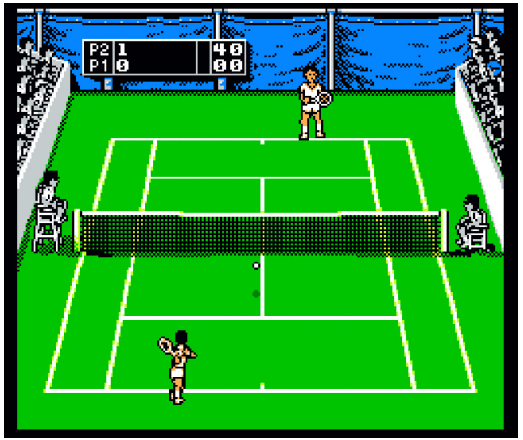
1 bit 2 bit 8 bit

8-Bit Colour

- Gives us 256 different colours
  - which 256 depends on the system

Computer Gaming Graphics (1)

- 1980s – Nintendo Entertainment System (NES)
  - 8 bit console
  - 54 colours



Computer Gaming Graphics (2)

- Early 1990s – Super Nintendo Entertainment System (SNES)
  - 16 bit console
  - 256 colours

Computer Gaming Graphics (3)

- Mid 1990s – Sony PlayStation
  - 32 bit console
  - 16,777,216 colours

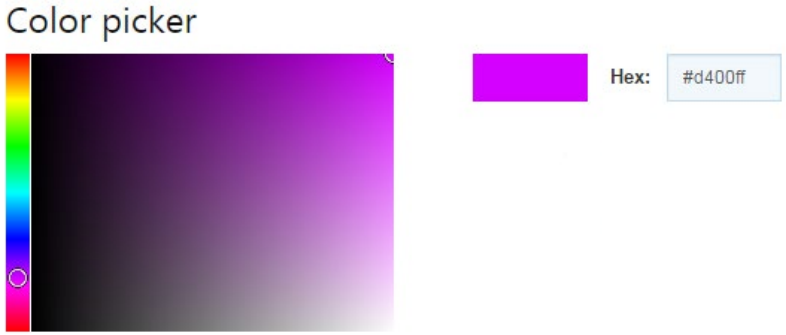
Hexadecimal

- Base 16 numbering system
  - 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F
- Represented with 0x__
  - i.e. 255 in decimal is 0xFF
- More readable than binary
  - decimal 121
  - binary 01111001
  - hex 0x79
- Stores more information than binary
  - hex and binary relate well
  - 1 hex digit can always be represented with 4 binary digits
  - no relationship like this between decimal and binary

decimal	hexadecimal	binary
0	0	0000
1	1	0001
2	2	0010
3	3	0011
4	4	0100
5	5	0101
6	6	0110
7	7	0111
8	8	1000
9	9	1001
10	A	1010
11	B	1011
12	C	1100
13	D	1101
14	E	1110
15	F	1111

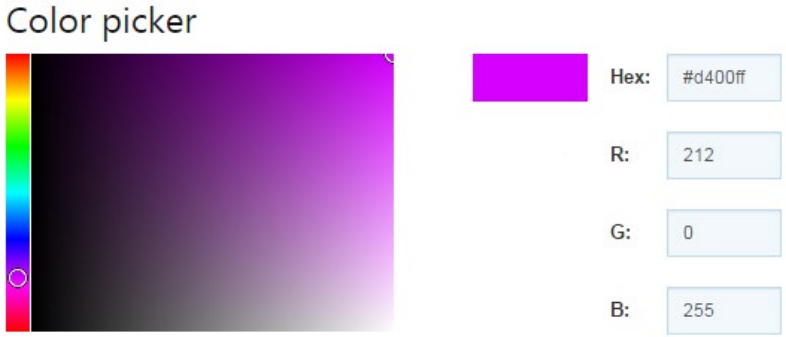
24-bit Colour (1)

- 24-bit colour means that there are 24 bits allocated for RGB channels
  - 8 bits for red, 8 bits for green, 8 bits for blue
- How to represent this colour purple?
  - rich purple? deep purple?
- Need a better representation
  - often use hexadecimal 0xd400ff
  - what does that mean?



24-bit Colour (2)

- Hex 0xd400ff
  - d4 – red bits
  - 00 – green bits
  - ff – blue bits



- Compact, as opposed to 24-bit value for this colour purple
  - 110101000000000011111111

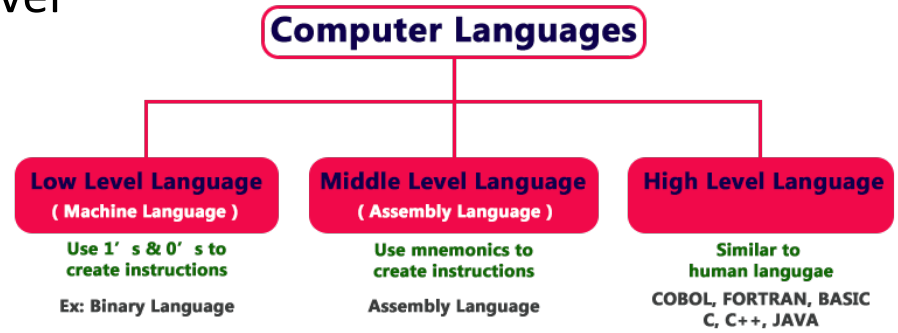
ASCII – American Standard Code for Information Interchange

- Recall, computers can only store binary
  - includes characters
- ASCII is an encoding scheme that represents each character with a 7-bit value

Dec	Hex	Oct	Char	Dec	Hex	Oct	Char	Dec	Hex	Oct	Char	Dec	Hex	Oct	Char
0	0	0		32	20	40	[space]	64	40	100	@	96	60	140	`
1	1	1		33	21	41	!	65	41	101	A	97	61	141	a
2	2	2		34	22	42	"	66	42	102	B	98	62	142	b
3	3	3		35	23	43	#	67	43	103	C	99	63	143	c
4	4	4		36	24	44	$	68	44	104	D	100	64	144	d
5	5	5		37	25	45	%	69	45	105	E	101	65	145	e
6	6	6		38	26	46	&	70	46	106	F	102	66	146	f
7	7	7		39	27	47	'	71	47	107	G	103	67	147	g
8	8	10		40	28	50	(	72	48	110	H	104	68	150	h
9	9	11		41	29	51	)	73	49	111	I	105	69	151	i
10	A	12		42	2A	52	*	74	4A	112	J	106	6A	152	j
11	B	13		43	2B	53	+	75	4B	113	K	107	6B	153	k
12	C	14		44	2C	54	,	76	4C	114	L	108	6C	154	l
13	D	15		45	2D	55	-	77	4D	115	M	109	6D	155	m
14	E	16		46	2E	56	.	78	4E	116	N	110	6E	156	n
15	F	17		47	2F	57	/	79	4F	117	O	111	6F	157	o
16	10	20		48	30	60	0	80	50	120	P	112	70	160	p
17	11	21		49	31	61	1	81	51	121	Q	113	71	161	q
18	12	22		50	32	62	2	82	52	122	R	114	72	162	r
19	13	23		51	33	63	3	83	53	123	S	115	73	163	s
20	14	24		52	34	64	4	84	54	124	T	116	74	164	t
21	15	25		53	35	65	5	85	55	125	U	117	75	165	u
22	16	26		54	36	66	6	86	56	126	V	118	76	166	v
23	17	27		55	37	67	7	87	57	127	W	119	77	167	w
24	18	30		56	38	70	8	88	58	130	X	120	78	170	x
25	19	31		57	39	71	9	89	59	131	Y	121	79	171	y
26	1A	32		58	3A	72	:	90	5A	132	Z	122	7A	172	z
27	1B	33		59	3B	73	;	91	5B	133	[	123	7B	173	{
28	1C	34		60	3C	74	<	92	5C	134	\	124	7C	174	|
29	1D	35		61	3D	75	=	93	5D	135	]	125	7D	175	}
30	1E	36		62	3E	76	>	94	5E	136	^	126	7E	176	~
31	1F	37		63	3F	77	?	95	5F	137	_	127	7F	177	

Computer Instructions

- Computers also use binary for their instructions!
- However, we do not program in binary
  - this would be a nightmare to write and proofread
  - so many errors
- Programming languages are developed to help this
  - they can be low-level, or high-level



Java

- Created by James Gosling
- Modern object-oriented language
  - continuously improved since 1990s
- Widely used
  - mars rover, cell phones, web servers, ...

- Unfortunately, Java is verbose
  - using more words – compare with **Python**!

- More importantly, it is not about the language
  - the programming skills that you learn in this course will apply to many languages

What Java can do

- **Android Apps**: Java is one of the primary languages for Android app development.

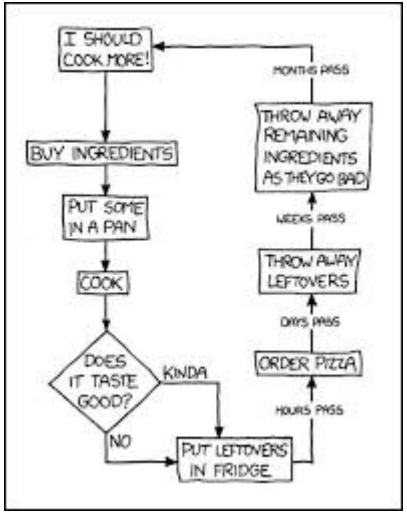
- **Minecraft**: The original version of this immensely popular game was developed in Java.

- **Spotify**: A popular music streaming service.

- **Eclipse IDE**: An integrated development environment used primarily for Java development.

Learn to Program

- At its core, a program is **a set of instructions** that a computer follows to perform **a specific task**.
  - Think of it like a **recipe**. Just as a recipe tells you step-by-step how to bake a cake, a program tells the computer step-by-step how to complete a task.
- Programming is the **process of writing these instructions**. It's like being a chef who creates recipes. In programming, you use a special language (called a programming language) to write these instructions so that the computer can understand and execute them.



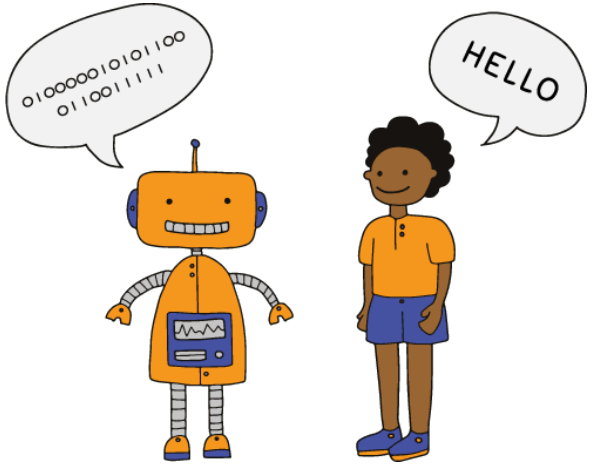
Programming Languange

- Imagine that you want to tell your friend how to bake a cake.
  - “First, mix the flour and sugar. Then, add eggs and milk. Finally, bake it in the oven.”
  - A programming language is similar, but instead of telling a friend how to bake a cake, you are telling a computer how to do a task.

- Just like human languages such as English, Spanish, or Chinese, there are many different programming languages, like Python, Java, and C++.

Human vs. Programming Language

- Vocabulary and Grammar:
  - Human Language: Words and rules (grammar) to form sentences. For example, “The cat is on the mat.”
  - Programming Language: Keywords and syntax (rules) to give instructions. For example, in Java, **System.out.println("Hello, World!")** tells the computer to display the text “Hello, World!”.
- Learning:
  - Human Language: You learn it by speaking, reading, and writing.
  - Programming Language: You learn it by writing code and running programs to see if they work.
- Translation:
  - Human Language: If you speak English and your friend speaks Spanish, you might need a translator.
  - Programming Language: Computers use special programs called **compilers** or **interpreters** to translate the code you write into a language the computer can understand.



First Program: Hello World

- Write a program that prints "`Hello World`" to the screen

First Program: Hello World

- Write a program that prints "Hello World" to the screen

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

- Let’s have a try
  - https://onecompiler.com/java

Integrated Development Environment (IDE)

- An **Integrated Development Environment (IDE)** is a software application that provides comprehensive facilities to computer programmers for software development. It typically includes:
  - **Source Code Editor**: A text editor designed specifically for writing and editing source code.
  - **Compiler/Interpreter**: A tool that translates the written code into a language that the computer can understand and execute
  - **Build Automation Tools**: Tools that automate the process of compiling and building the software, saving time and reducing the risk of human error.
  - **Debugger**: A tool that helps you test and debug your programs by allowing you to run your code step-by-step, inspect variables, and track down errors.

Integrated Development Environment (IDE)

- Visual Studio Code https://code.visualstudio.com/
  - A lightweight, open-source IDE from Microsoft, popular for its wide range of extensions and support for multiple programming languages.
- Eclipse https://www.eclipse.org/downloads/
  - A versatile, open-source IDE primarily used for Java development but also supports other languages through plugins.
- IntelliJ IDEA https://www.jetbrains.com/idea/
  - A robust IDE from JetBrains, known for its advanced code analysis and powerful refactoring tools, mainly used for Java development.
- Netbeans https://netbeans.apache.org/front/main/index.html
  - NetBeans IDE is a free and open source integrated development environment for application development.

First Program: Hello World

- Let's start from here to show what is problem-solving and how to learn programming

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

- Which part is the easiest to understand?
- And what is the key part in commanding?
- Then is there any parts that are similar?

First Program: Hello World

The Java main program skeleton:

package name

```
package xjtlu.cpt111.introduction;
```

All opening brackets need to be matched!!

class name

```
public class HelloWorld {
```

main() method

```
  public static void main(String[] arguemnts) {
    // display message "Hello World" on the console
    System.out.println("Hello World!");
  }
```

Method block

code to execute

```
}
```

Class block

Object-Oriented Language

- Java is an Object-Oriented Language:
  - every Java file must contain a *class declaration*
  - all code is written inside a *class*
  - to run a Java program, need to define a *main method*

more OOP features in future lectures...

- Java's language features:
  - braces/curly brackets are used to denote the beginning and end of a block/a section of code
    - e.g. a class or method declaration
  - statements end with semicolons

- Violating one of the above will cause *syntax errors*

Elements of the Hello World Program

- What are the elements of the Hello World program?

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

class

main method

parameter

statement

braces括号

squared brackets方括号

parentheses花括号

semicolon分号

Double quotes in println()

- What will happen if we remove the double quotes around `"Hello World"` ?

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello World");
    }
}
```

- Then, what if we want to print exactly `"Hello World"` (instead `Hello World`) in the output?

Escape character in println()

- Strings must be written within quotes, thus we need an **escape character (\)** to tell the program that it is for output.

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello World");
        System.out.println("\"Hello World\"");
    }
}
```

- `\" -> "`
- `\' -> '`
- `\\ -> \`

Comments in Java

- Comments are used to contain non-coding contents in programming

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello World");
// this will make a single line to be skipped
/* this will make the
   whole section to be skipped
*/
        System.out.println("\"Hello World\"");
    }
}
```

- Use comments to leave notes and change the program in running

Variable

- Write a program that prints an integer variable num which stores an integer value 5

```
public class Hello {
    public static void main(String[] args) {
        int num = 5;
        System.out.println(num);
    }
}
```

variable

type

value

visualize the code execution in https://cscircles.cemc.uwaterloo.ca/java_visualize/

add some integer to the variable num before printing the result

expression

Static Typing

- Java is a statically typed language:
  - all variables, parameters, and methods must have a *declared type*
    - also called their *static type*
  - variables must be declared before use
  - that type can never change
  - expressions also have a type, e.g. `5 + 10` has type `int`
- Java compiler ensures type consistency
  - if types are inconsistent, the program will not compile
    - it is another example of *syntax error*
- Using an IDE, the compiler checks that all the types in your program are compatible *before* the program ever runs
  - a good thing, catch errors as early as possible

In-Class Quiz

- What is the output of the following program?

```
public class Hello {
    public static void main(String[] args) {
        int num = 5;
        num = num + num;
        System.out.println(num);
    }
}
```

- 0
- 5
- 10
- 55

what if you want to reuse the initial value of `num` in other computation?

Declaration, Initialization, Assignment

- Variables can be declared and initialized once, and assigned multiple times

```
public class Hello {
    public static void main(String[] args) {
        int num1 = 5;
        int num2;
        num2 = 10;
        int num = num1 + num2;
        System.out.println(num);
    }
}
```

declaration

initialization

assignment

Thank you for your attention !

- In this lecture, you have learned:
  - how to compile and run a simple Java program
  - about the structure of a simple Java program
  - how to use variables to store data